Online Supplemental Appendix

Electroencephalography Educational Initiatives for Non-experts in the Acute Care Environment: A Systematic Review and Meta-analysis

Shaurya Taran MD, Wael Ahmed MD, Ruxandra Pinto, Esther Bui MD, Lara Prisco MD MSc AFRCA AFFICM, Cecil D. Hahn MD MPH, Marina Englesakis BA, Victoria A. McCredie MBChB PhD

eAppendix 1: MEDLINE Search Strategy

eAppendix 2: Data Extraction Form

eAppendix 3: List of Authors Contacted

eAppendix 4: NIH Quality assessment and study adjudication

eAppendix 5: Breakdown of EEG Studies by Theme

eAppendix 6: Detailed Summary of EEG Studies

eAppendix 7: Teaching Methods in EEG Studies

eAppendix 8: Funnel Plot for Pre-Post Studies

eAppendix 9: Funnel Plot for Kappa Studies

eAppendix 10: Risk of Bias Assessment for Randomized Control Trials

eAPPENDIX 1: Search Strategy for MEDLINE

The following tables record the search strategies and terms used in MEDLINE. Search results were obtained from the inception of each of the databases to the third week of January 2020. An update of the search using the same strategy was performed shortly before submission for publication.

#	Searches	Results
1	exp electroencephalography/ or exp brain waves/ or alpha rhythm/ or beta rhythm/ or delta rhythm/ or gamma rhythm/ or theta rhythm/ or electrocorticography/ or exp electroencephalography phase synchronization/ or cortical synchronization/ or magnetoencephalography/	155887
2	electroencephalogra*.mp,jw.	156747
3	electro-encephalogra*.mp.	806
4	diagnostic techniques, neurological/ and (exp Brain/ or exp Brain Diseases/ or exp Brain Injuries/)	510
5	brain wave?.mp.	4062
6	alpha rhythm?.mp.	4944
7	beta rhythm?.mp.	2038
8	delta rhythm?.mp.	1612
9	gamma rhythm?.mp.	1019
10	theta rhythm?.mp.	5014
11	electrocorticogra*.mp.	3566
12	electro-corticogra*.mp.	30
13	(cortical adj1 synchroni*).mp.	3112
14	magnetoencephalogra*.mp.	8665
15	magneto-encephalogra*.mp.	149
16	continuous EEG?.mp.	784
17	aeeg.ti,ab,kf.	388
18	ceeg.ti,ab,kf.	266
19	cvEEG?.ti,ab,kf.	10
20	iEEG?.ti,ab,kf.	335
21	microEEG?.mp.	4
22	micro-EEG?.mp.	14
23	mEEG?.ti,ab,kf.	14
24	quantitative EEG?.mp.	1230
25	QEEG?.ti,ab,kf.	864
26	spot EEG?.mp.	2
27	sEEG?.ti,ab,kf.	584

28	videoelectroencephalogra*.mp.	20
29	video-electroencephalogra*.mp.	955
30	videoEEG?.mp.	25
31	video-EEG?.mp.	2566
32	vEEG?.ti,ab,kf.	236
33	or/1-32 [Electroencephalography]	170920
34	exp Intensive Care Units/	81164
35	exp Critical Care/	56212
36	Critical illness/	27534
37	Critical Care Nursing/	1909
38	critical care.mp.	64323
39	(critical?? adj2 ill*).mp,kw.	50446
40	intensive care?.mp.	147568
41	neurocritical care?.mp.	1054
42	neuro-critical care?.mp.	43
43	neurointensiv*.mp.	698
44	neuro-intensiv*.mp.	184
45	"burn department*".mp.	48
46	"burn unit?".mp.	3289
47	"burn ward?".mp.	108
48	"coronary care department*".mp.	8
49	"coronary care unit?".mp.	6828
50	"coronary care ward?".mp.	6
51	(CICU or CICUs).mp.	182
52	(CVICU or CVICUs).mp.	49
53	(high* adj2 depend* adj2 department*).mp.	6
54	(high* adj2 depend* adj2 unit*).mp.	578
55	(high* adj2 depend* adj2 ward*).mp.	37
56	(ICU or ICUs).mp.	46307
57	(intens* adj2 therap* adj2 department*).mp.	45
58	(intens* adj2 therap* adj2 unit*).mp.	753
59	(intens* adj2 therap* adj2 ward*).mp.	37
60	(medic* adj2 surg* adj2 ICU?).mp.	1059
61	"med* surg* icu?".mp.	570
62	medsurg icu?.mp.	0

63	MSICU.mp.	9
64	(NICU or NICUs).mp.	8496
65	(post-an?esth* adj2 care*).mp.	915
66	(postan?esth* adj2 care*).mp.	1983
67	"recovery department*".mp.	13
68	"recovery unit*".mp.	354
69	"recovery ward*".mp.	103
70	(surg* adj2 med* ICU?).mp.	79
71	exp Respiration, Artificial/	74978
72	ventilat*.mp.	165475
73	exp Ventilators, Mechanical/	8953
74	(an?esthe* adj2 recover*).mp.	7262
75	Anesthesia Recovery Period/	5115
76	"Delayed Emergence from Anesthesia"/	117
77	*Acute Disease/	8253
78	*Catastrophic Illness/	568
79	acute care unit?.mp.	372
80	acute care ward?.mp.	175
81	acute care department?.mp.	16
82	Tertiary Care Centers/	12317
83	(tertiary adj1 care).mp.	43514
84	exp Emergency Medicine/	13354
85	Emergencies/	39760
86	exp Emergency Medical Services/	135395
87	Emergency Nursing/	6987
88	exp Emergency Service, Hospital/	74908
89	Emergency Treatment/	10493
90	exp Evidence-Based Emergency Medicine/	395
91	Hospital Rapid Response Team/	733
92	Pediatric Emergency Medicine/	226
93	"accident? and emergenc*".mp.	4303
94	emergency center?.mp.	571
95	emergency centre?.mp.	153
96	emergency department?.mp.	71676
97	emergency room?.mp.	15546

98	emergency service?.mp.	71053
99	emergency unit?.mp.	1823
100	emergency visit?.mp.	1071
101	emergency ward?.mp.	899
102	or/34-101 [Critical Care / Intensive Care / Emergency Department]	637190
103	33 and 102 [Electroencephalography + ICU/CC]	4817
104	"Facility Regulation And Control"/	3174
105	"Joint Commission On Accreditation Of Healthcare Organizations"/	7449
106	Clinical Competence/	90819
107	Ed.fs. [Education Floating Subheading]	272723
108	Education, Medical, Continuing/	24542
109	exp Academic Medical Centers/	91562
110	exp Accreditation/	18648
111	exp Certification/	17973
112	exp Competency-Based Education/	3860
113	exp Consensus Development Conference/	11749
114	exp Consensus Development Conferences As Topic/	2787
115	exp Consensus/	11956
116	exp Credentialing/	54028
117	exp Curriculum/	83134
118	exp Education, Medical, Graduate/	67694
119	exp Education, Medical/	161582
120	exp Education, Professional/	292966
121	exp Educational Measurement/	144883
122	Evaluation Studies/	247852
123	Evaluation Studies as Topic/	121674
124	exp Faculty, Medical/	12922
125	exp Faculty/	34601
126	Feasibility Studies/	65397
127	exp Hospitals, Teaching/	50409
128	exp Inservice Training/	28657
129	exp Interdisciplinary Studies/	944
130	exp International Educational Exchange/	3458
131	exp Licensure/	17386
132	exp Medicine/ [Specialties]	1099637

133	exp Mentors/	10533
134	exp Models, Educational/	9942
135	exp Physicians/	135789
136	exp Preceptorship/	4999
137	exp Professional Competence/	114052
138	exp Schools, Medical/	25198
139	exp Teaching Materials/	117468
140	exp Teaching/	83838
141	Hospitals, Teaching/	22521
142	Hospitals, University/	29195
143	Inservice Training/	20020
144	Pilot Projects/	118638
145	Professional Competence/	24004
146	Program Development/	28424
147	Program Evaluation/	61429
148	Specialty Boards/	3627
149	exp Video Recording/	40138
150	st.fs. [Standards]	701845
151	Training Support/	6051
152	(evaluat* adj2 (study or studies)).mp.	519204
153	(feasibilit* adj2 (study or studies)).mp.	70323
154	(fellow or fellows or fellowship*).mp,kw.	24125
155	(intern or interns or internship?).mp.	52554
156	(resident or residents or residency or residencies).mp.	161544
157	(skill or skilled or skills).mp,kw.	181559
158	(training? or trainee?).mp,kw.	366386
159	(video* adj2 record*).mp.	42358
160	accredit*.mp,kw.	29996
161	academ*.mp,kw.	156637
162	certif*.mp,kw.	56122
163	competen*.mp,kw.	208604
164	(course or courses).mp.	533839
165	credential*.mp,kw.	6754
166	curricul*.mp,kw.	91523
167	curricula?.mp,kw.	15878

168	educat*.mp,kw.	915067
169	governance.mp,kw.	10234
170	instruction.mp,kw.	34951
171	learn*.mp,kw.	387698
172	licenc*.mp,kw.	2243
173	licens*.mp,kw.	43741
174	mentor*.mp,kw.	18017
175	non-neuro*.mp.	9074
176	nonneuro*.mp.	3094
177	non-specialist??.mp.	1522
178	nonspecialist??.mp.	610
179	non-technologist?.mp.	0
180	nontechnologist?.mp.	4
181	outline?.mp,kw.	96109
182	postgrad*.mp,kw.	14639
183	post-grad*.mp,kw.	2409
184	preceptor*.mp,kw.	6452
185	proctor*.mp,kw.	710
186	(program? or programme?).mp,kw.	811798
187	qualif*.mp,kw.	41447
188	seminar?.mp,kw.	7983
189	standard?.mp,kw.	1456337
190	syllab*.mp,kw.	6990
191	train*.mp,kw.	459917
192	tutor*.mp.	8547
193	videorecord*.mp.	428
194	video-record*.mp.	29051
195	workshop*.mp,kw.	32089
196	or/104-195 [Education & related terms]	5504883
197	103 and 196 [Eelectroencephalography + ICU/CC + Education]	1455
198	exp animals/ not (exp animals/ and exp humans/)	4663383
199	197 not 198	1383
200	limit 197 to humans	1375
201	199 or 200	1383
202	remove duplicates from 201	1378

eAPPENDIX 2: Data Extraction Form for Eligible Studies

Reviewer: _____ Date of Extraction: _____
(dd/mm/yyyy)

Article Identification:

RefWorks ID#: _____
Title:
_____
_____
_____

First Author: _____
Senior Author: _____
Journal Name/Reference Source: _____

Publication type:
- ☐ Journal Article
- ☐ Published Abstract
- ☐ Conference Proceeding
- ☐ Other: _____

Year: _____ Volume: _____ Starting Page: _____

Full Publication: ☐ Yes ☐ No Comment: _____

Language of Publication: English Other: _____

Source of sponsorship (Study): _____
Study – Country of Origin: _____
Number of Centers Participating: _____
Authors contacted for additional data ☐ Yes ☐ No

Verification of Eligibility:
Inclusion Criteria: (verified by "tick" mark)

- ☐ Case series, descriptive studies, cohort study OR Randomized Control Trial
- ☐ Adult OR pediatric patients received EEG (either intermittent/short EEG, video EEG, continuous EEG (cEEG), processed EEG (e.g., BIS, entropy) or quantitative EEG (QEEG)) for a pre-specified indication (e.g. seizure diagnosis)
- ☐ Study was conducted in one of the following inpatient settings: emergency department (ED), intensive care unit (ICU), or post anesthetic care unit (PACU)
- ☐ Study described training/educational programs for interpretation of EEG (either raw OR processed data)
- ☐ Participants did not have prior expertise in EEG interpretation
- ☐ Study reported sufficient details/data of the training program, including but not limited to, structure, content covered, duration, assessment methods, and trainee feedback

ONLY for assessment of secondary outcomes, include the following:

- ☐ Studies provided sufficient data to calculate performance pre- and post- training program, AND/OR inter-rater agreement coefficient with experienced cEEG readers

Study Design:

Type of Study:

- ☐ Randomized controlled trial
- ☐ Retrospective cohort study
- ☐ Prospective cohort study
- ☐ Pre-post interventional study
- ☐ Cross-sectional study
- ☐ Case series
- ☐ Descriptive study

Inclusion Criteria:

1. _____
2. _____
3. _____
4. _____
5. _____

Exclusion Criteria:

1. _____
2. _____
3. _____
4. _____
5. _____

Screening and Enrollment:

<u>For RCTs and Cohort studies</u>

Total participants enrolled: _____ Control pts: _____ Intervention pts: _____

Timing of enrolment (mm/yy): Control pts: _____ Intervention pts: _____
Consecutive participants: ☐ Yes ☐ No ☐ Not specified
Description of why not: _____

Participant identification:

- ☐ ICU database
- ☐ Prospective enrolment
- ☐ Other: _____
- ☐ Hospital database
- ☐ Medical records coding

For all other studies

Total participants enrolled: _____

Timing of enrolment (mm/yy):

Consecutive participants: ☐ Yes ☐ No ☐ Not specified

Description of why not: _____

Study setting:

- ☐ ED
- ☐ ICU (describe specific setting, e.g. MS-ICU, TN-ICU): _____
- ☐ PACU
- ☐ Mixed setting: _____

Study Participant Characteristics:

Non-expert review by:

- ☐ Intensive care physicians
- ☐ Neurologists (NOT epileptologists/neurophysiologists)
- ☐ ED physicians
- ☐ Anesthesiologists
- ☐ Nurses
- ☐ Other: _____
- ☐ Mixed group: _____

Median years in practice (if reported): _____

Involvement of trainees:

- ☐ Yes (describe level and background): _____
- ☐ No
- ☐ Not specified

Number of non-expert participants: _____

Expert review by:

- ☐ Neurophysiology technologist/ Electroencephalographer
- ☐ Neurophysiology doctors
- ☐ Neurology nurses
- ☐ Neurologist with specialization in epileptology
- ☐ Other: _____
- ☐ Mixed group: _____

Number of experts performing gold-standard interpretations: _____

Patient Characteristics:

- ☐ Adult (age > 18 yrs)
- ☐ Pediatric (age < 18 yrs)
- ☐ Neonate (age < 4 wks)
- ☐ Mixed group: _____

Clinical diagnosis on admission:

Neurologic

- ☐ Seizure
- ☐ Traumatic brain injury
- ☐ Subarachnoid hemorrhage
- ☐ Stroke
  i. Hypertensive
  ii. Ischemic
- ☐ Hypoxic ischemic encephalopathy (e.g. post cardiac arrest)
- ☐ Other: _____

Non-neurologic

- ☐ Sepsis
- ☐ Acute respiratory distress syndrome (ARDS)
- ☐ Trauma
- ☐ Other: _____

Aspects of EEG recordings:

- ☐ Short/intermittent EEG (raw)
- ☐ Continuous EEG (cEEG) (i.e., >4 hours recordings)
- ☐ Quantitative EEG (QEEG)
- ☐ Simplified processed EEG (Sedline, Bis, Entropy, CSA, etc)
- ☐ Other: _____

Clinical indications for EEG recording:

- ☐ Suspected non-convulsive status epileptics (NCSE)
- ☐ Convulsive status epilepticus
- ☐ Monitoring treatment response for seizures
- ☐ Monitoring depth of sedation
- ☐ Abnormal movements of unclear etiology
- ☐ Altered mental status NYD

- ☐ Toxic/metabolic encephalopathy
- ☐ Not reported

Total hours of cEEG recorded: _____

Total number of cEEGs obtained: _____

EEG setup:

- ☐ Full montage (i.e., 21 electrode)
- ☐ Simplified full-scalp montage (i.e., 8-16 electrodes)
- ☐ Limited montage (e.g., 2 or 4 electrodes limited to one scalp region)
- ☐ Not reported

Training program characteristics:

Did program involve the following:

- ☐ PowerPoint
- ☐ Self-study
- ☐ Simulations
- ☐ Bedside teaching

Description of training program (describe specific methods, duration, content covered, and any other relevant aspects:

_____
_____
_____
_____
_____

Description of assessment strategy (e.g., written quiz, pass/fail, score-based):

_____
_____
_____

Passing mark (if score-based): _____

Retention strategy (if described):

_____
_____

Trainee feedback (if reported): _____
_____

Cost of program: _____

Patient Baseline Characteristics*:

	Control (n =)	**Intervention (n =)**
Demographics		
Age (SD)		
Male Sex		
APACHE (SD)		
SOFA/MODS/SAPS		
Admission GCS (SD)		
Post-op admission (Y/N)		
Transferred from another institution (Y/N)		
Admission data		
Patients with status epilepticus (%)		
Patients with pre-existing epilepsy (%)		
Post-cardiac arrest (%)		
Primary diagnosis was stroke (%)		
Primary diagnosis was TBI (%%		
Primary diagnosis was SAH (%)		
Primary diagnosis was non-neurologic (e.g., sepsis, ARDS, %)		
Baseline Hemodynamics		
BP		
HR		
T_{max}		
GCS		
SpO2		

*For case series, omit “control” arm. Leave fields unfilled if data was not reported.

Primary Outcome:

Provide qualitative summary and description of educational program: _____
_____
_____
_____

Secondary Outcomes

For pre-post-studies*:

Variable	Pre-program	Post-program
Overall score		
% answers correct		
Sensitivity		
Specificity		
PPV		
NPV		
p-value		
Background activity		
% answers correct		
Sensitivity		
Specificity		
PPV		
NPV		
p-value		
Artefact		
% answers correct		
Sensitivity		
Specificity		
PPV		
NPV		
p-value		
Sleep-wake cycle		
% answers correct		

Sensitivity		
Specificity		
PPV		
NPV		
p-value		
Pathologic trace (specify which type if available)		
% answers correct		
Sensitivity		
Specificity		
PPV		
NPV		
p-value		
Burst suppression		
% answers correct		
Sensitivity		
Specificity		
PPV		
NPV		
p-value		
Seizure detection		
% answers correct		
Sensitivity		
Specificity		
PPV		
NPV		
p-value		

*Fill table if study assessed performance of participants on assessments before and after completion of the training program.

For interpretation against gold standard*

Variable	Trainee
Background activity	
% answers correct	
Sensitivity	
Specificity	
PPV	
NPV	
Inter-rater agreement coefficient	
Strength of agreement	
Artefact	
% answers correct	
Sensitivity	
Specificity	
PPV	
NPV	
Inter-rater agreement coefficient	
Strength of agreement	
Sleep-wake cycle	
% answers correct	
Sensitivity	
Specificity	
PPV	
NPV	
Inter-rater agreement coefficient	
Strength of agreement	
Pathologic trace (specify which type if available)	
% answers correct	

Sensitivity	
Specificity	
PPV	
NPV	
Inter-rater agreement coefficient	
Strength of agreement	
Burst suppression	
% answers correct	
Sensitivity	
Specificity	
PPV	
NPV	
Inter-rater agreement coefficient	
Strength of agreement	
Seizure detection	
% answers correct	
Sensitivity	
Specificity	
PPV	
NPV	
Inter-rater agreement coefficient	
Strength of agreement	

*Fill table if study assessed performance of participants compared to a “gold-standard” reader (e.g. neurophysiologist/epileptologist

eAPPENDIX 3: List of Authors Contacted

Author(s): Amorim
Paper: Performance of spectrogram-based seizure identification of adult EEGs by critical care nurses and neurophysiologists
Reason for contacting: Data request for secondary outcome analysis
Author Response: Yes. Data provided.

Author(s): Bouwens
Paper: Training neonatal nursing staff in continuous EEG in a Level III NICU
Reason for contacting: Abstract with incomplete data
Author Response: No. Study was not included due to limited description of training program.

Author: De Vries
Paper: Neuromonitoring; how to train your nursing staff
Reason for contacting: Abstract with incomplete data
Author Response: Yes. Study was not included after clarification of information with author.

Author(s): Fahy/Bensalem-Owen
Paper: The effectiveness of a simple novel approach on EEG instruction for anesthesiology residents
Reason for contacting: Material potentially duplicated in another study
Author Response: Yes. Study was not included after clarification of information with author.

Author(s): Fantaneanu
Paper: Resident training and interrater agreements using the ACNS critical care EEG terminology
Reason for contacting: Some results of interest but unclear from the manuscript
Author Response: Yes. Study was not included after clarification of information with author.

Author(s): Kang
Paper: A trial of real-time electrographic seizure detection by neuro-ICU nurses using a panel of quantitative EEG trends
Reason for contacting: Data request for secondary outcome analysis
Author Response: Yes. Data provided.

Author(s): Kromm/Kramer
Paper: Development of an EEG curriculum for icu nurses to facilitate real time screening of continuous EEG data for seizures in critically ill adults
Reason for contacting: Abstract with incomplete data
Author Response: Yes. Study was not included after clarification of information with author.

Author(s): Leira
Paper: Continuous or emergent EEG: can bedside caregivers recognize epileptiform discharges?
Reason for contacting: Data request for secondary outcome analysis
Author Response: Yes. Data provided.

Author(s): Parvizi
Paper: Individuals without EEG training can detect subclinical seizures and seizure-like activity by listening to sonified EEG
Reason for contacting: Abstract with incomplete data
Author Response: Yes. Study was not included after complete review.

Author(s): Poon
Paper: Integrated approach to e-learning enhanced both subjective and objective knowledge of aEEG in a neonatal intensive care unit
Reason for contacting: Data request for secondary outcome analysis
Author Response: Yes. Data provided.

Author(s): Seiler
Paper: The effectiveness of a staff education program on the use of continuous EEG with patients in neuroscience intensive care units
Reason for contacting: Data request for secondary outcome analysis
Author Response: Yes. Data provided.

Author(s): Topjian
Paper: Detection of electrographic seizures by critical care providers using color density spectral array after cardiac arrest is feasible
Reason for contacting: Data request for secondary outcome analysis
Author Response: Yes. Data provided.

eAPPENDIX 4: National Institute of Health (NIH) Quality Assessment Tools

Supplemental Appendix 4a: Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies

For each question, mark "yes", "no", or "N/A"

1. Was the research question or objective in this paper clearly stated?
2. Was the study population clearly specified and defined?
3. Was the participation rate of eligible persons at least 50%?
4. **Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study prespecified and applied uniformly to all participants?**
5. Was a sample size justification, power description, or variance and effect estimates provided?
6. For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?
7. Was the timeframe sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?
8. For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome (e.g., categories of exposure, or exposure measured as continuous variable)?
9. **Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?**
10. Was the exposure(s) assessed more than once over time?
11. **Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?**
12. Were the outcome assessors blinded to the exposure status of participants?
13. Was loss to follow-up after baseline 20% or less?
14. Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?

Overall quality rating (Good, Fair, or Poor)

If Poor, please state why:

Reviewer 1 initials
Reviewer 2 initials

eAPPENDIX 4b: Quality Assessment Tool for Before-After (Pre-Post) Studies With No Control Group

For each question, mark “yes”, “no”, or “N/A”

1. Was the study question or objective clearly stated?
2. Were eligibility/selection criteria for the study population prespecified and clearly described?
3. Were the participants in the study representative of those who would be eligible for the test/service/intervention in the general or clinical population of interest?
4. Were all eligible participants that met the prespecified entry criteria enrolled?
5. Was the sample size sufficiently large to provide confidence in the findings?
6. **Was the test/service/intervention clearly described and delivered consistently across the study population?**
7. **Were the outcome measures prespecified, clearly defined, valid, reliable, and assessed consistently across all study participants?**
8. Were the people assessing the outcomes blinded to the participants' exposures/interventions?
9. Was the loss to follow-up after baseline 20% or less? Were those lost to follow-up accounted for in the analysis?
10. Did the statistical methods examine changes in outcome measures from before to after the intervention? Were statistical tests done that provided p values for the pre-to-post changes?
11. **Were outcome measures of interest taken multiple times before the intervention and multiple times after the intervention (i.e., did they use an interrupted time-series design)?**
12. If the intervention was conducted at a group level (e.g., a whole hospital, a community, etc.) did the statistical analysis take into account the use of individual-level data to determine effects at the group level?

Overall quality rating (Good, Fair, or Poor)

If Poor, please state why:

Reviewer 1 initials
Reviewer 2 initials

eAPPENDIX 4c: Process of adjudicating study quality

Following the Agency for Healthcare Research and Quality (AHQR) recommendations, the overall quality of studies as "good," "fair," or "poor" was adjudicated using a set of rules. With rules-based assignment, the AHQR recommends selecting a list of items from the study tool that capture important domains of a study's internal validity. If studies meet the majority of items across the domain, they may be assigned a "good" rating; those that meet some of the domains are "fair;" and all others are poor. Full details of the AHQR recommendations are available at the following link: https://effectivehealthcare.ahrq.gov/products/methods-guidance-bias-individual-studies/methods.

For cohort studies, question items 4, 9, and 11 of the NIH quality assessment tool (bolded in Supplemental Appendix 4a) were considered to be prerequisites for strong internal validity. Studies meeting all 3 questions were assigned a quality rating of "good;" those meeting one or two questions were assigned a "fair" rating, and those meeting none were assigned a "poor" rating.

For pre-post studies, question items 6, 7, and 11 of the NIH quality assessment tool (bolded in Supplemental Appendix 4b) were considered to be prerequisites for strong internal validity. Studies meeting all 3 questions were assigned a quality rating of "good;" those meeting one or two questions were assigned a "fair" rating, and those meeting none were assigned a "poor" rating.

Differences in quality assessment between the two primary reviewers were resolved in consensus.

eAPPENDIX 5: Additional Study Characteristics

Features	Number	Studies by Primary Author
Study Design		
Cohort	26	Amorim, Bourgoin, Chau, Citerio, Dericioglu, Du Pont-Thibodeau, Fahy (2008), Fahy (2009), Fahy (2019), Fahy (2020), Ganesan, Goswami, Griesmaier, Guan, Herta, Kang, Kyriakopoulos, Legriel, Leira, Lybeck, Nitzschke, Poon, Rennie, Seiler, Swisher, Topjian
Descriptive	2	Sacco, Whitelaw
RCT	1	Chari
Publication year		
2016 – 2020	16	Amorim, Bourgoin, Chari, Citerio, Du Pont-Thibodeau, Fahy (2019), Fahy (2020), Ganesan, Goswami, Guan, Herta, Kang, Kyriakopoulos, Legriel, Lybeck, Sacco
2006 - 2015	11	Chau, Dericioglu, Fahy (2008), Fahy (2009), Griesmaier, Nitzschke, Poon, Seiler, Swisher, Topjian, Whitelaw
Before 2005	2	Leira, Rennie
Place of the study		
Asia	2	Guan, Poon
Europe	10	Bourgoin, Citerio, Dericioglu, Griesmaier, Herta, Legriel, Lybeck, Nitzschke, Rennie, Whitelaw
North America	17	Amorim, Chari, Chau, Du Pont-Thibodeau, Fahy (2008), Fahy (2009), Fahy (2019), Fahy (2020), Ganesan, Goswami, Kang, Kyriakopoulos, Leira, Sacco, Seiler, Swisher, Topjian
Number of centers		
1	22	Bourgoin, Chau, Dericioglu, Du Pont-Thibodeau, Fahy (2008), Fahy (2009), Fahy (2019), Fahy (2020), Ganesan, Guan, Kang, Kyriakopoulos, Leira, Lybeck, Nitzschke, Poon, Rennie, Sacco, Seiler, Swisher, Topjian, Whitelaw
2	5	Amorim, Citerio, Goswami, Griesmaier, Herta
3 or more	2	Chari, Legriel
Study setting		
Adult ICU	8	Herta, Kyriakopoulos, Legriel, Leira, Lybeck, Nitzschke, Swisher, Topjian
Adult Neuro-ICU	10	Amorim, Chau, Citerio, Dericioglu, Fahy (2008), Fahy (2009), Fahy (2019), Fahy (2020), Kang, Seiler
Pediatric ICU	4	Bourgoin, Du Pont-Thibodeau, Ganesan, Guan
Neonatal ICU	6	Goswami, Griesmaier, Poon, Rennie, Sacco, Whitelaw
ED	1	Chari
Participant background		
Anesthesiology residents	3	Fahy (2008), Fahy (2009), Fahy (2019)

ED physicians	1	Chari
ICU physicians	11	Citerio, Du Pont-Thibodeau, Griesmaier, Guan, Legriel, Lybeck, Nitzschke, Poon, Rennie, Topjian, Whitelaw
ICU fellows	7	Chau, Du Pont-Thibodeau, Fahy (2020), Ganesan, Legriel, Leira, Topjian
ICU nurses	17	Amorim, Bourgoin, Dericioglu, Du Pont-Thibodeau, Ganesan, Goswami, Griesmaier, Guan, Herta, Kang, Legriel, Leira, Poon, Sacco, Seiler, Swisher, Topjian
Neurology residents	3	Dericioglu, Kyriakopoulos, Leira
Neurosurgery residents	1	Leira
Other residents	2	Griesmaier, Legriel
Other participants	6	Ganesan, Guan, Herta, Legriel, Leira, Swisher
Number of participants		
< 19	12	Chau, Dericioglu, Fahy (2009), Fahy (2020), Ganesan, Guan, Herta, Kyriakopoulos, Lybeck, Nitzschke, Rennie, Swisher
20 - 49	8	Amorim, Chari, Du Pont-Thibodeau, Fahy (2008), Kang, Poon, Seiler, Topjian
>50	7	Bourgoin, Citerio, Fahy (2019), Goswami, Griesmaier, Legriel, Leira
N/A	2	Sacco, Whitelaw
EEG focus		
aEEG	15	Bourgoin, Dericioglu, Du Pont-Thibodeau, Ganesan, Goswami, Griesmaier, Guan, Herta, Kang, Lybeck, Nitzschke, Poon, Rennie, Sacco, Swisher
CDSA	6	Amorim, Dericioglu, Du Pont-Thibodeau, Ganesan, Swisher, Topjian
cEEG	7	Chau, Citerio, Goswami, Legriel, Lybeck, Seiler, Whitelaw
Intermittent EEG	7	Chari, Fahy (2008), Fahy (2009), Fahy (2019), Fahy (2020), Kyriakopoulos, Leira
Other	3	Herta, Kang, Topjian
Teaching methods		
Bedside/Interactive	13	Chau, Fahy (2008), Fahy (2009), Ganesan, Goswami, Guan, Kang, Kyriakopoulos, Lybeck, Nitzschke, Rennie, Seiler, Whitelaw
Case-based learning	5	Citerio, Fahy (2020), Legriel, Lybeck, Whitelaw
Didactic lectures	19	Chari, Chau, Citerio, Du Pont-Thibodeau, Fahy (2008), Fahy (2019), Griesmaier, Guan, Herta, Kyriakopoulos, Legriel, Leira, Lybeck, Nitzschke, Poon, Sacco, Swisher, Topjian, Whitelaw
One-on-one review	7	Chau, Dericioglu, Fahy (2008), Fahy (2009), Ganesan, Nitzschke, Rennie
Online modules	5	Bourgoin, Fahy (2020), Goswami, Kyriakopoulos, Poon
PowerPoint	9	Chari, Du Pont-Thibodeau, Ganesan, Kang, Leira, Poon, Seiler, Swisher, Topjian
Question and answer	7	Du Pont-Thibodeau, Kyriakopoulos, Nitzschke, Rennie, Sacco, Topjian, Whitelaw
Simulation	4	Fahy (2020), Goswami, Sacco, Whitelaw
Small group discussion	5	Amorim, Bourgoin, Dericioglu, Poon, Seiler
Textbook readings	1	Guan
Assessment methods*		

Real-time/bedside interpretation	9	Bourgoin, Dericioglu, Guan, Kang, Kyriakopoulos, Lybeck, Nitzschke, Rennie, Swisher
Written quiz	18	Amorim, Chari, Chau, Du Pont-Thibodeau, Fahy (2008), Fahy (2009), Fahy (2019), Fahy (2020), Ganesan, Griesmaier, Herta, Legriel, Leira, Poon, Sacco, Seiler, Topjian, Whitelaw
Duration of learning program*		
< 3 hours	8	Amorim, Du Pont-Thibodeau, Ganesan, Herta, Kang, Legriel, Nitzschke, Swisher
3-6 hrs	6	Bourgoin, Chau, Dericioglu, Griesmaier, Poon, Rennie
7 hrs – 1 day	3	Lybeck, Sacco, Whitelaw
>1 day	4	Fahy (2008), Fahy (2009), Fahy (2019), Guan

*Only for studies reporting this data

aEEG amplitude integrated electroencephalography, *cEEG* continuous electroencephalography, *CDSA* color density spectral array, *ED* emergency department, *ICU* intensive care unit, *RCT* randomized control trial

Author (year)	Study objectives	Structure of EEG Training Program	Evaluation methods	Main Findings
Amorim (2017)	To evaluate the performance characteristics of CSA-based seizure detection by neurocritical care nurses after a brief training program.	All nurses participated in a small group session for 15 minutes. Tutorial content included CSA theory, CSA interpretation and correlation with raw EEG data, and epileptic patterns and artifact identification using CSA. The tutorial was followed by a web-based test, with a 10-question pretest, and a discussion of pitfalls on interpretation of CSA images for 25 minutes.	A test with 40 different CSA images was performed after the tutorial, and discussion of pretest answers was completed. For each CSA image, participants were asked to indicate whether any seizures were present and seizure count.	Average seizure detection sensitivity for nurses was 73.8%, with a specificity of 37.6%. The false alarm rate was one per 3.2 hours. The overall accuracy for the presence or absence of seizures was 55.7% for nurses and 67.5% for neurophysiologists. Seizure identification by ICU nurses using CSAs after a brief training session appeared to be feasible.
Bourgoin (2019)	To evaluate the ability of bedside nurses to detect abnormal traces after receiving aEEG training. Performance was compared to EEG interpretation by neurophysiologists.	Training included a 45-minute web-based learning on basics of electroencephalography and aEEG, standardized electrodes placement, pattern recognition, seizure detection, and artifacts analysis. After completion of this web-based learning, participants were divided into small groups and a debriefing was conducted.	Participants completed trace analysis forms. Participants had to recognize and allocate background aEEG patterns as moderately altered, burst suppression, flat trace, or seizures. Results were compared with those obtained by neurophysiologists.	The agreement between nurses and experts was moderate for the recognition of any pathologic pattern (k=0.53, 95% CI: 0.39–0.67). Recognition of a severely altered background resulted in a better nurse/ expert concordance (k=0.71, 95% CI: 0.57–0.85). Finally, moderate agreement was found for the detection of seizures only (k = 0.40, 95% CI: 0.25–0.54).
Chari (2019)	To describe the utility of a brief training module for enhancing ED physicians' ability to identify seizures by intermittent EEG.	Participants were randomized to the intervention group or the control group. Those in the intervention group received a self-learning PowerPoint presentation covering the basic principles of EEG, including EEG recording techniques, montages, and characteristics of normal and abnormal patterns. The control group was asked to take the quiz without reviewing the training slides.	Participants completed a 20-question quiz. Each test EEG was accompanied by two questions: normal or abnormal, and seizure vs no seizure. The EEG quiz consisted of normal samples and abnormal samples showing focal or generalized seizures, slowing, burst suppression, spikes, and triphasic waves (1)). The quiz was repeated 2 months later to test retention.	The percentage of correct answers in the intervention group (65%, quartiles 63% and 75%) was significantly different (p = 0.001) from that of control group (50%, quartiles 46% and 59%) for the initial quiz. For the 2-month follow-up retention quiz, the intervention group performed better than the control group (68% [quartiles 60% and 73%] versus 58% [quartiles 55% and 61%]) but the difference was not statistically significant (p = 0.05).
Chau (2014)	To understand the effectiveness of an interdisciplinary electroencephalogram (EEG) educational module for critical care trainees.	The training curriculum covered the basics of monitoring, physiologic basis, and the relevant clinical applications of EEG. The residents also interpreted 10 EEGs with a neurophysiologist (1:1 faculty-to-fellow ratio). The total time spent in this EEG learning module during the NSICU month was 4 hours per fellow. Bedside learning also	Participants completed the baseline evaluation tool before any EEG instruction, followed by another assessment when 10 EEG interpretations with the neurophysiologist were completed. Each evaluation tool contained 25 multiple-choice items. One point was	The mean test score increased from 7.56 ± 2.24 before the learning module to 16.67 ± 2.96 (p < 0.001) after the module. All participants increased their scores compared with baseline after the EEG learning module.

		occurred in the form of cEEG review in critically ill patients admitted to the ICU.	assigned to each correct answer for a total maximum grading score of 25 points.	
Citerio (2017)	To train intensivists in interpreting cEEG and qEEG, and to evaluate the agreement between intensivists and neurophysiologists in interpreting relevant information on cEEG/qEEG	The intensivists received a short neurophysiology training course consisting of lectures on cEEG and qEEG interpretation given by neurophysiologists. Interactive with the neurophysiologist was also provided.	Intensivists were evaluated on their ability (1) to assess the depth of sedation, (2) to evaluate symmetry between the hemispheres, (3) to recognize seizures, and (4) to recognize artefacts. Their responses were compared to those of two experienced neurophysiologists, used as the "gold standard".	Following the training program, there was improvement in all of the domains explored (symmetry, sedation, artefacts, seizures detection), with the most improvement in the detection of artefacts and seizures. The detection of symmetry and sedation were already acceptable in the before-training period and did not improve substantially with training.
Dericioglu (2015)	To investigate the diagnostic utility of aEEG and CDSA by non-expert physicians and ICU nurses for NCSE identification in adult patients	All participants received group and individual training sessions for a total of 5-6 hours per participant. The training material consisted of theoretical knowledge and examples of aEEG and CDSA traces with and without seizures.	The performance of raters for identifying the presence or absence of seizures was assessed for each time epoch consisting of 1 h-long aEEG/CDSA recordings. True positives, false positives, true negatives, and false negatives were calculated by comparison to EEG interpretation by an epileptologist.	Compared to gold standard reading by an epileptologist, aEEG and CDSA provided high sensitivity (overall 93%), specificity (91-95%) and accuracy (AUC 0.93) for seizure detection in the hands of ICU personnel with no experience in critical care EEG and epilepsy.
Du Pont-Thibodeau (2017)	To determine the accuracy of critical care medicine providers to identify seizures using aEEG versus aEEG + CDSA.	Each participant received a 30-minute standardized slide tutorial. The tutorial presented 1) basics of cEEG, 2) basics of aEEG and CDSA interpretation, 3) seizure detection examples, and 4) difficult pattern examples including artifacts such as movement artifact. The participants were allowed to ask questions and review the tutorial for as long as needed.	Participants took a quiz classifying aEEG and aEEG + CDSA image as having seizure(s) or no seizure. Each participant reviewed 200 images. Each aEEG image question was followed by the same aEEG image combined with the matching CDSA image. Participants were asked to determine whether seizures were present or absent on each image.	After a brief training program, pediatric ICU providers were able to achieve a moderate sensitivity and high NPV for seizure detection using either aEEG or aEEG + CDSA. Sensitivity and NPV using aEEG were 77% and 88%, respectively, and 78% and 89% using aEEG + CDSA.
Fahy (2008)	To test the effectiveness of a novel educational program for EEG interpretation by anesthesiology residents.	The EEG curriculum included didactic lectures on EEG principles interspersed with operating room and ICU EEG exposure. An epileptologist interpreted EEG studies obtained throughout the institution twice weekly with the anesthesiology residents.	Participants completed a 25-item multiple choice quiz. The quiz included 11 EEG tracings for interpretation and questions relevant to intraoperative EEG interpretation for anesthesiologists. An independent third party graded the evaluation results in blinded fashion.	The educational effort resulted in significant improvement in test scores. For the individuals (n = 7) with evaluations before and after the education EEG intervention, mean scores improved from 10.7 ± 3.9 to 18.8 ± 3.0 ($p = 0.001$) with the new educational EEG exposure.

Fahy (2009)	To increase familiarity with EEG use and interpretation in residents completing their neurosurgical ICU rotation. .	Residents accompanied the clinical neurophysiologist to the operating room for EEG interpretation. One-on-one instruction was provided during a wide range of neurosurgical cases.	The residents were evaluated with four evaluation tools, each containing 25 multiple choice items developed by the investigators. The quiz was taken before and after participation in the training program.	The mean score increased from 8.00 ± 2.51 before the EEG interpretations to 15.12 ± 3.00 ($p < 0.001$), 15.88 ± 3.18 ($p = 0.001$), and 18.12 ± 3.23 ($P = 0.001$) after interpreting 10, 15, and 20 EEGs, respectively.
Fahy (2019)	To assess the effectiveness of an EEG training module on anesthesiology residents' performance on in-training evaluations (ITEs).	The module consisted of didactic material encompassing general principles of EEG, components required for monitoring, recognition of normal/abnormal traces, and specialized application of EEG to other settings (e.g. coma, brain death, and cerebral ischemia).	Residents undertook completion of the annual in-training examination (ITE), which assesses basic EEG competencies required of anesthesiology trainees. The ITE is offered by the American Board of Anesthesiology.	Residents who received the educational module showed greater improvements compared with corresponding residents who did not receive the educational module.
Fahy (2020)	To improve EEG interpretation as assessed by evaluation tools before and after participants received a "flipped classroom" based EEG curriculum.	The EEG curriculum covered principles of technology and monitoring, adult normal awake and sleep patterns, and abnormal EEG patterns. A flipped classroom approach was used, whereby students performed self-directed learning at home, followed by neurophysiologist-scheduled instructional EEG reading session ("classroom" time).	Evaluations took place using an interactive EEG simulator which displayed EEG tracings from simulated clinical encounters. Participants had to answer questions related to each EEG. The evaluation had 25 questions and was performed before and after participants received the training program.	Participants showed significant improvement in their EEG interpretation skills ($p = 0.008$). Mean EEG evaluation tool scores increased from baseline (10.8 ± 2.9) following the simulation curriculum (15.6 ± 3.2). Overall satisfaction with the program was moderately high (mean score 3.7 ± 0.9), with 5 being highest satisfaction level.
Ganesan (2018)	To compare the performance of critical care providers with that of EEG experts in identifying seizures using qEEG.	Participants underwent standardized one-on-one training, composed of a 30-minute slide-based presentation introducing the theoretical basis and practical use of two qEEG tools (aEEG and CDSA), followed by 90 minutes of hands-on training using qEEG review software to examine EEG recordings.	All participants were evaluated on their ability to identify seizures in a set of 27 cEEG recordings while viewing only the qEEG trends. They were given written instructions to mark any epochs that they "suspected to be seizures." Sensitivity, daily false-positive rates, and positive predictive values were calculated by comparing seizures marked by participants with those that had previously been identified on the raw EEG by a board-certified neurophysiologist.	Following 2 hours of training, critical care providers were able to use qEEG displays to detect at least one seizure in 94% of critically ill children who underwent cEEG monitoring and were able to identify individual seizures with a median sensitivity of 76.9% using aEEG and 85.4% using CDSA. The performance of critical care providers was comparable with EEG experts in terms of their sensitivity, positive predictive values, and false-positive rates for seizure identification.
Goswami (2018)	To establish consistency in the monitoring of newborns admitted to 2 different	The educational sessions focused on lead application, impedance checks, camera position, initiation and stopping a recording, and troubleshooting equipment issues. Monthly skills sessions were also held on	No evaluation methods were used.	The time to initiate brain monitoring decreased by 31.5 hours post training; this, in turn, led to an increase in electrographic seizure detection (20% before vs 34% after), decrease in seizure clinical

	NICUs following an educational program.	seizure causes and the pathophysiology of common conditions such as HIE.		misdiagnosis (65% before and 36% after), and reduction in antiseizure medication burden.
Griesmaier (2011)	To investigate the effects of a brief training program on the ability of non-specialists to diagnose seizure activity.	Physicians received special aEEG training for 60 min three times per year. Technical aspects of EEG, underlying physiology, normal and abnormal background patterns, and disturbances in aEEG recordings caused by external factors were discussed during training.	Physicians completed a questionnaire that was composed of 16 questions and divided into two parts: The first part asked questions about the daily use of aEEG. The second part asked the staff to interpret aEEG traces that were obtained from patients.	Seizure activity was correctly diagnosed by 33 respondents (60.0%). Subgroup analyses showed that 11 (84.6%) of the 13 physicians and 22 (52.4%) of the 42 nurses made the correct diagnosis. Flat trace and sleep–wake cycling were both diagnosed correctly by 61.8% of respondents.
Guan (2016)	To evaluate the feasibility of independent use of aEEG by pediatric ICU providers to identify seizures in critically ill children.	Participants received training from an experienced epileptologist over a period of 4 hrs. Training was based on principles from the textbook "Newborn amplitude integrated EEG atlas." The training content included the basic principles of aEEG, various background graphics, identification of onset graphic features and artifacts, etc. Participants then conducted 15 reading exercises of various types of aEEG.	Following the training, subjects independently interpreted aEEGs without receiving additional clinical information. Participants were asked to mark the time point and type of the seizures on the time axis of the aEEG trend graph. The original vEEG was interpreted by a physician and constituted the gold-standard reading.	Each subject had the highest rate of correct recognition of status epilepticus. The average correct recognition rate for transient seizures was 66 ± 4%, and the missed rate was 24.1% -32.6%.
Herta (2017)	To assess whether ICU caregivers can correctly read and interpret cEEG data displayed with the computer algorithm NeuroTrend (NT), with the main objective of detecting seizures and depth of sedation.	All respondents underwent a brief educational course lasting approximately one hour. Features of NTs graphical user interface and the study design were explained. Samples of cEEG and NT data were presented.	Each respondent was asked to rate the preselected NT data. The rating manual could be used during the evaluation process. During the assessment, the time needed to evaluate the NT data of 20 patients was measured.	Perfect agreement was found for spike waves, rhythmic delta activity, and burst suppression. Electrographic seizure patterns, periodic discharges. and seizure suspicion achieved substantial agreement among the respondents. Similar agreements were found when the choices of the respondents were compared with expert opinion.
Kang (2019)	To evaluate the performance of Neuro-ICU nurses utilizing bedside, real-time qEEG interpretation for detecting recurrent NCSE.	Consented Neuro-ICU nurses underwent a 15-min qEEG training via a Powerpoint presentation. A 1-h qEEG panel printout containing the patient's most recent seizure(s) was displayed next to the bedside cEEG/qEEG acquisition monitor, and nurses were instructed to observe for similar patterns.	For the duration of their shifts, the nurses logged the number of seizures seen hourly based on their qEEG interpretation. The corresponding conventional cEEG segments were reviewed independently by the study authors.	With only brief training, Neuro-ICU nurses were able to detect the presence of recurrent electrographic seizures using a simplified panel of qEEG trends with good sensitivity and specificity (85.1% and 89.8%, respectively).

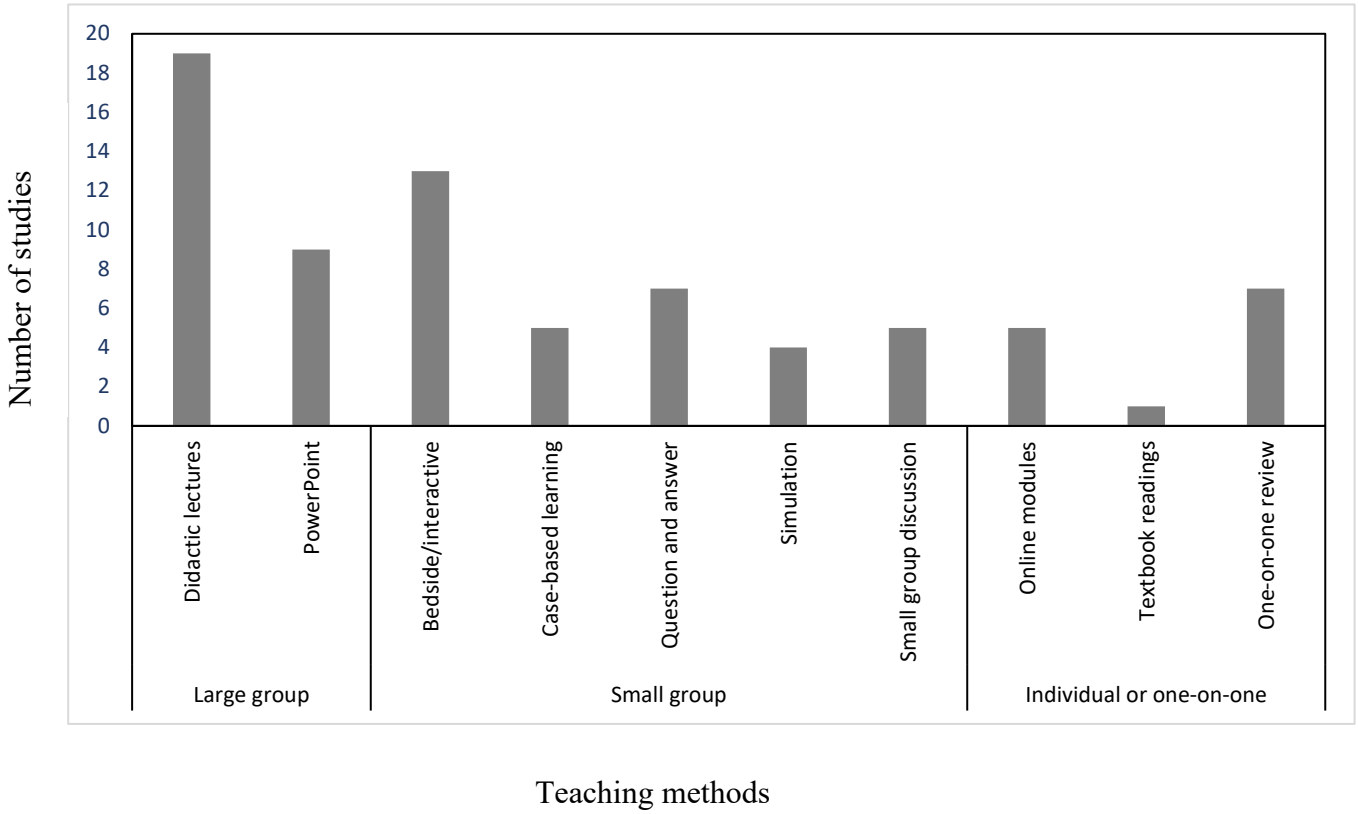
Kyriakopoulos (2020)	To test whether neurology residents can be trained to use and interpret full-montage EEGs using an EEG cap electrode system to detect NCSE while on-call.	EEG training for participants consisted of self-study of articles and standardized training slides retrieved from the ACNS website. Participants subsequently reviewed the training slides with an epilepsy specialist during a 2-hour didactic teaching period. All residents additionally received a 2-hour training session on the application of an EEG cap system	Neurology residents interpreted tracings obtained from cap EEGs during their on-call shifts. Results were compared to "gold-standard" interpretation of the same tracings by 3 epilepsy specialists. Kappa coefficients were calculated for EEG pattern identification.	Residents were capable of capturing highly interpretable records; however, they only interpreted them 57% of the time in line with the gold standard. Residents did not miss any malignant patterns concerning for NCSE but "overcalled" patterns as malignant in 3 of 14 recordings.
Legriel (2020)	To evaluate the effectiveness of a training program on basic interpretation of cEEG for critical care staff in a prospective multicenter study.	Participants were given a 90-min face to-face EEG course, consisting of theoretical concepts and illustrative EEG traces. Traces were selected to be illustrative of the ten predefined EEG learning objectives covering the most common normal and abnormal patterns encountered in critically ill patients. Additionally, participants were given a pocket flyer and notebook summarizing the key points of the face-to-face EEG course.	Participants were evaluated before receiving the training program and at multiple follow-up time points, up to day 90. For each evaluation, participants received an online survey consisting of ten questions: seven simple choice questions (one item from a list of five choices) and three multiple choice questions (one or more items from a list of five choices). All ten questions allowed evaluation of the ten points of the EEG learning objectives at each survey. The order of the questions and the corresponding proposed answers were chosen randomly.	In total, 77/108 (71.3%) trainees achieved at least 80% correct answers in the day-90 evaluation after the face to-face EEG course.
Leira (2004)	To assess the ability of caregivers to recognize epileptiform discharges, and to determine whether skill improves with a simple education program.	Participants received a 40-slide PowerPoint lecture aimed to improve recognition of epileptiform discharges. The lecture covered basic aspects of EEG recording, including normal EEG tracings, artifacts, and epileptiform discharges.	Participants received a pre-test followed by the instructional PowerPoint, followed by the post-test. The pre- and post- tests consisted of 24 EEG images accompanied by the following question: "Is this EEG most consistent with epileptiform discharges?"	The mean number of correct responses in the overall population was 14.72 (61%) on the pre-test and increased to 16.06 (67%) on the post-test (p=0.002). Performance was exceptionally good for the EEG technicians, and reasonably good for neurology residents.
Lybeck (2019)	To assess ICU physicians' ability to perform preliminary interpretations of a simplified cEEG in the post cardiac arrest setting.	The training consisted of a half-day educational session and a follow-up training assignment, for a total training duration of one day. The educational session was held by two neurophysiologists and consisted of lectures, case-based discussions and interpretation of cEEG recordings with help from neurophysiologists.	The evaluation consisted of 10 cEEG recordings from a separate cohort and a one-hour video tutorial with detailed explanation of the neurophysiologist's interpretation of the practice cEEGs. The ICU physicians' interpretations were compared to those of an experienced interpreter.	There was substantial strength of agreement (Kappa = 0.69) for EEG background. Identification of a continuous EEG background also yielded substantial agreement (K = 0.69) and was detected with median sensitivity 86% (95% CI 71%-95%) and specificity 74% (95% CI 57%-88%). There was moderate agreement (K = 0.43) for classification of epileptiform discharges.

Nitzschke (2011)	To evaluate the diagnostic use of aEEG for the detection of seizures in adult patients by physicians with no prior aEEG exposure.	Participants received two training sessions on an individual basis. Interactive cases were used to explain core concepts of EEG. Each observer was allowed to ask as many questions as required in order to understand the interpretative methods. The time allotted to each reviewer was 1 h for each training session on two different days.	After training, reviewers evaluated the aEEG traces to distinguish the presence or absence of seizures. All observers were blinded to the raw EEG trace and to the clinical data of the patient. The reviewers had no time constraints, and each reviewer had full access to their training material for reference during the evaluation.	Single channel aEEG recordings were not reliable for the identification of convulsive generalized epileptic seizures when reviewed by nonexpert ICU physicians. False-negatives rates were high and accordingly the sensitivity was low, with a median sensitivity across all observers of 40%.
Poon (2015)	To evaluate the objective and subjective effectiveness of an integrated e-learning module for aEEG in the NICU.	The e-learning program consisted of three modules of online aEEG teaching lasting 1 hour each, administered over a period of 3 months. The modules covered: (a) the basics of aEEG; (b) the fundamentals of aEEG interpretation; and (c) recognition of seizures. Protected time was given for the online teaching sessions, which were followed by facilitated discussions, and questions and answer periods.	Participants completed a 25-question survey, of which five questions tested the participant's ability to correctly identify aEEG tracings. These five tracings were chosen from among six possible aEEG pattern traces (continuous normal voltage, discontinuous normal voltage, continuous low voltage, burst suppression, inactive and seizure activity).	A total of 77.8% (21/27) of participants felt that the duration of the teaching was optimal, 96.2% (25/26) felt that the teaching was well structured, 80.0% (20/25) felt that they had learned a systematic approach to interpreting aEEGs, and 70.4% (19/27) felt that they could interpret normal aEEGs with confidence. When the pre- and post- program results were compared, there was a statistically significant improvement in the participants' subjective knowledge score with median improvement from pre- to post-programme score ($p = 0.041$).
Rennie (2004)	To determine the accuracy of non-expert use of the CFM and to assess interobserver agreement of CFM seizure detection.	All participants received training sessions on an individual basis. The process of CFM interpretation was explained to each observer using a module containing 20 interpreted CFM traces with and without seizures. Each observer was allowed to ask as many questions as required on interpretative methods. The time allotted to each observer was generally between three and five hours.	Each participant evaluated 120 anonymous CFM traces, 19 from seizure cases and 21 from controls. The participants had no time constraints, and each had full access to their training material for reference during the evaluation.	CFM was not reliable for the diagnosis of seizure when interpreted by nonexpert users. When results were compared with cEEG readings, a large number of seizures were missed.
Sacco (2016)	To present a broad educational program for neonatal nurses caring for patients undergoing therapeutic hypothermia.	The training module was divided into didactic material (3 hours) and simulations (2 hours). Topics covered in the didactic session included pathophysiology of HIE, principles of aEEG, and basic strip evaluation. Participants next visited the simulation lab	A follow-up survey was completed once the program concluded to assess the nurse's opinion of the program's effectiveness. The survey included objective questions about EEG to test participants' acquisition of the material.	The educational components of this program demonstrated improved learner knowledge, with pretest scores averaging 60 percent and posttest scores averaging 95 percent. Course evaluations were overall positive.

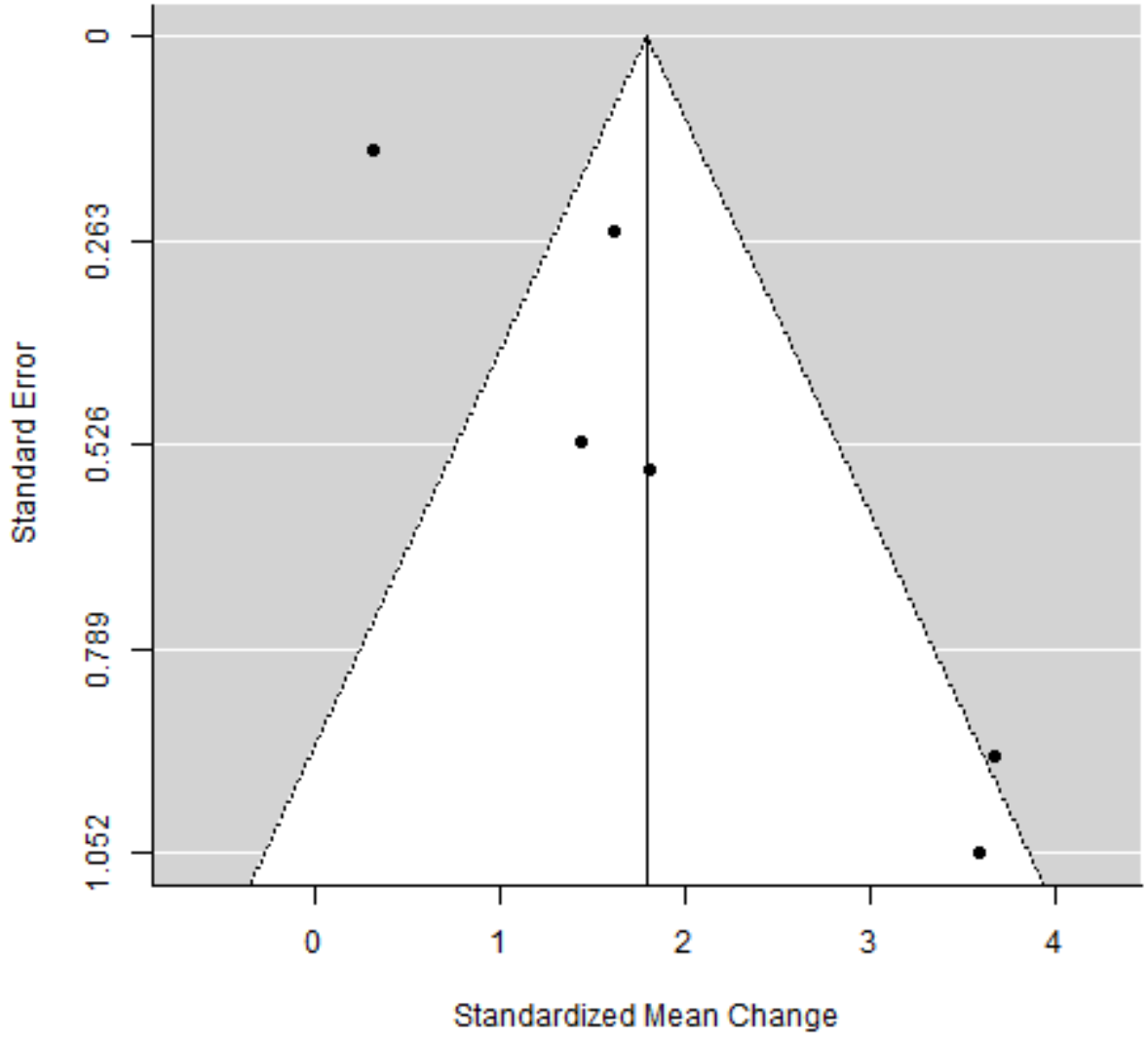
		to practice their knowledge and skills in an observed environment. .		
Seiler (2012)	To evaluate the effectiveness of a staff educational program aimed at improving the knowledge of nurses in the use of cEEG monitoring in adults.	The educational program included both a lecture/discussion component and a hands-on clinical component. A PowerPoint presentation covering the theory of EEG was followed by a simulation session in which participants visited the EEG department to practice lead application and use of machines.	A 20-question multiple choice, true/false test was administered to participants before and after completion of the training program. In addition, a 1-month follow-up test was performed to test retention of the subject matter.	On the pretest, more than half of the participants scored below 14 (57.4%), with a range of scores from 4 to 18 and a mean score of 12.7. On the posttest, 100% had a score of 14 or higher. On the 1-month follow-up test, 90% had a score of 14 or better, representing a significant improvement from pretest to posttest.
Swisher (2015)	To evaluate the sensitivity and specificity of a panel of qEEG trends for seizure detection in adult ICU patients when reviewed by neurophysiologists and non-neurophysiologists.	Participants were given a brief 15-minute PowerPoint presentation on qEEG. The presentation consisted of a basic description of each of the qEEG trends used in the study, examples of seizures on qEEG panels and several slides contrasting episodes of artifact from seizures.	A total of 180 randomized qEEG panels were distributed to participants. All reviewers analyzed all 180 qEEG panels. These reviewers were asked to determine the number of seizures present in the panel with the options being: no seizures, 1 to 2 seizures, 3 to 5 seizures, 6 to 10 seizures or 10 seizures.	For the reviewer's ability to detect the presence of seizures on qEEG panels when compared with the gold standard of independent raw EEG review, the sensitivities and specificities were as follows: neurophysiologists 0.87 and 0.61, EEG technologists 0.80 and 0.80, and Neuroscience ICU nurses 0.87 and 0.61, respectively.
Topjian (2015)	To determine the accuracy and reliability of seizure detection by critical care providers using CDSA.	Each participant received a 15-minute PowerPoint slide tutorial addressing (1) EEG basics, (2) description of CDSA components, (3) interpretation of CDSA, (4) recognition of seizures on CDSA, and (5) recognition of challenging CDSA patterns such as artifacts and burst suppression. The participants had as much time as needed to review the tutorial and ask questions.	Each participant was presented with a total of 200 CDSA images in random order. Participants were instructed to answer "yes" if they thought there was at least one seizure on the image or "no" if they thought there were no seizures on the image. The images were displayed and answers collected using survey monkey. Participants could not ask questions or receive feedback as they reviewed and scored the images during the testing.	Critical care providers had a CDSA seizure detection sensitivity of 70% [95% CI: 67%, 73%], specificity of 68% [95% CI: 67%, 70%], PPV of 46% and NPV of 86%. The Cohen's Kappa was 63% [95% CI: 61%, 66%] for all images and 69% [64%, 72%] for images with seizures. There was no difference between critical care attendings (AUC 0.68) and critical care nurses (AUC 0.66), but critical care fellows were better than both groups AUC 0.73 ($p<0.001$).
Whitelaw (2006)	To enable staff with little or no previous experience to initiate and interpret basic cEEG tracings. .	A 1-day course provided participants with a general overview of EEG concepts using a multi-modal approach. Lectures, illustrative cases, role playing, and practical sessions covered various aspects of EEG: electrode application, physiologic basis of recordings, seizure detection, and recognition of artefacts.	A quiz with 10 CFM/EEG traces with specific questions on the findings was administered to participants. This was marked and feedback on the correct answers was given.	A short, one-day course on the basics of cEEG may be effective at imparting important principles to non-experts.

ACNS American clinical neurophysiology society, *aEEG* amplitude integrated electroencephalography, *AUC* area under the curve, *CDSA* color density spectral array, *CSA* compressed spectral array, *cEEG* continuous electroencephalography, *CFM* cerebral function monitor, *ED* emergency department, *EEG* electroencephalography, *HIE* hypoxic ischemic encephalopathy, *ICU* intensive care unit, *ITE* in-training evaluation, *NCSE* nonconvulsive status epilepticus, *NICU* neonatal intensive care unit, *NPV* negative predictive value, *NSICU* neurosurgical intensive care unit, *NT* neurotrend, *PPV* positive predictive value, *qEEG* quantitative electroencephalography, *vEEG* video electroencephalography

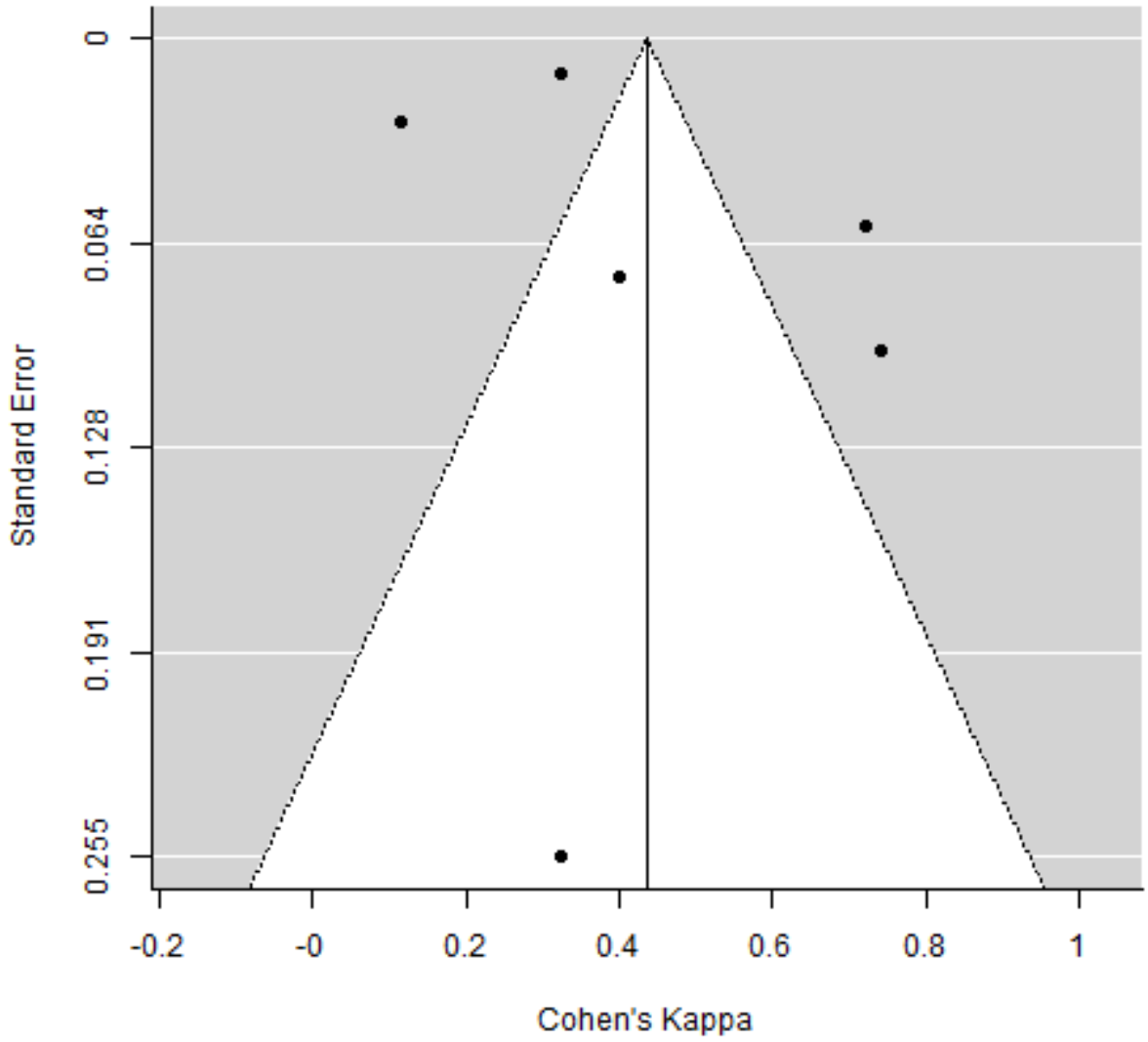
eAPPENDIX 7: Teaching Methods in EEG Studies



eAPPENDIX 8: Funnel Plot for Pre-Post Studies



eAPPENDIX 9: Funnel Plot for Kappa Studies



eAPPENDIX 10: Risk of Bias for Randomized Control Trials

One randomized control trial (RCT) was included in this systematic review. Risk of bias was assessed with the Cochrane Collaboration's Risk of Bias 2 (RoB 2) tool. The full RoB 2 checklist can be accessed from https://www.bmj.com/content/366/bmj.l4898.

